

BASIC OF ARTIFICIAL INTELLIGENCE

Complete In-Depth & Exam-Oriented Study Notes

B.Tech (Aeronautical Engineering)

Semester VI

Open Elective-II

Course Code: 100615

How to Use These Notes

These notes follow the supplied syllabus for Basic of Artificial Intelligence, B.Tech (Aeronautical Engineering), Semester VI, Course Code 100615, Open Elective-II. The supplied specification requires complete coverage with definitions, detailed explanations, working, examples, advantages, disadvantages, applications, comparisons, formulas, pseudocode, diagrams and exam questions.

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Study cycle: understand the definition → learn the working → reproduce the diagram/table → solve one example → practise a 2-mark and a long-answer question.

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Final Exam Preparation and Revision

UNIT 1 — INTRODUCTION TO AI & SEARCHING TECHNIQUES

This unit is expanded from every topic listed in the supplied syllabus. Each topic is presented with definition, simple explanation, working or key structure, example, exam point and relevant applications.

1. Introduction to Artificial Intelligence

Definition: AI is the field of computing concerned with systems that perceive information, reason, learn, solve problems and act toward goals.

In simple words: AI is broader than machine learning. It includes search, logic, knowledge representation, planning, language, decision making and learning. A useful agent view is percepts → internal processing → action.

Detailed explanation: A navigation system receives a destination and road information, searches possible routes and selects one according to a cost or preference.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

2. Types of AI

Definition: AI is commonly classified by capability and by functionality.

In simple words: Capability: Narrow/Weak AI is designed for a limited task; General/Strong AI is a theoretical goal of broad human-level flexibility; Super AI is hypothetical intelligence beyond humans. Functionality: reactive machines, limited-memory systems, theory-of-mind and self-aware categories.

Detailed explanation: A recommendation model is a narrow/task-specific system. General AI and Super AI should be treated as conceptual categories, not ordinary claims about current systems.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

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Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

3. Intelligent Agents

Definition: An agent perceives an environment through sensors and acts through actuators. An intelligent agent chooses actions appropriate to its objective.

In simple words: Agent = sense → decide → act. A percept is an observation; a percept sequence is the history of observations. A rational agent chooses an action expected to maximize its performance measure given available information and resources.

Detailed explanation: Vacuum agent: sensor detects dirt/location; actuator moves or cleans.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

4. PEAS and Task Environment

Definition: PEAS means Performance measure, Environment, Actuators and Sensors.

In simple words: Performance says how success is measured; Environment is the world; Actuators perform actions; Sensors provide observations. Environments can be fully/partially observable, deterministic/stochastic, episodic/sequential, static/dynamic, discrete/continuous, single/multi-agent and known/unknown.

Detailed explanation: Autonomous taxi: performance=safety/time/cost; environment=roads/traffic; actuators=steering/brakes; sensors=cameras/GPS.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

5. Types of Intelligent Agents

Definition: Main introductory architectures are simple reflex, model-based reflex, goal-based, utility-based and learning agents.

In simple words: Simple reflex uses current percepts and rules; model-based keeps internal state; goal-based reasons toward desired states; utility-based compares outcomes; learning agents improve from experience.

Detailed explanation: A thermostat is close to simple rule-based behavior; a route planner is goal-based; an adaptive recommender can be learning-based.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

6. State-Space Search

Definition: A state space represents possible configurations of a problem, with actions/operators connecting states.

In simple words: A search problem normally has an initial state, actions, transition model, goal test and path cost. A solution is an action sequence from start to goal.

Detailed explanation: Route finding: states=cities, actions=roads, goal=destination, path cost=distance/time.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

7. Breadth-First Search (BFS)

Definition: BFS expands the shallowest unexpanded node first using a FIFO queue.

In simple words: It explores level by level. It is complete for finite branching and optimal when step costs are equal. Main weakness is memory.

Detailed explanation: $A \rightarrow \{B, C\}$; $B \rightarrow \{D, E\}$; $C \rightarrow \{F, G\}$. BFS: A, B, C, D, E, F, G.

Example: Algorithm: 1) enqueue start; 2) dequeue front; 3) goal test; 4) enqueue successors; 5) repeat. Complexity: standard tree-search time/space are $O(b^{(d+1)})$ where d is shallowest goal depth.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

8. Depth-First Search (DFS)

Definition: DFS expands the deepest available node first using a stack or recursion.

In simple words: It follows one branch deeply, then backtracks. It uses less memory than BFS in many tree-search settings but is not generally optimal and can be incomplete in infinite spaces.

Detailed explanation: $A \rightarrow \{B, C\}$; $B \rightarrow \{D, E\}$; $C \rightarrow \{F, G\}$. Left-to-right DFS: A,B,D,E,C,F,G.

Example: Algorithm: push start; pop top; goal test; push successors; repeat. Typical tree-search time $O(b^m)$, space $O(bm)$, where m is maximum depth.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

9. Hill Climbing

Definition: Hill climbing repeatedly moves from the current state to a better neighboring state.

In simple words: Simple hill climbing takes an improving successor; steepest-ascent checks all neighbors and takes the best; stochastic chooses among improvements probabilistically. Problems include local maxima, plateaus and ridges.

Detailed explanation: In scheduling, a neighbor may be produced by swapping two jobs. Random restart can help escape local maxima.

Example: Key remedies: random restart, limited sideways moves, stochastic selection or a different global-search method.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

10. Heuristic Search and Best-First Search

Definition: A heuristic $h(n)$ estimates remaining cost or desirability. Best-first search chooses the frontier node with the best evaluation value.

In simple words: Greedy best-first commonly uses $f(n)=h(n)$, so it prefers nodes that look close to the goal but ignores cost already paid.

Detailed explanation: If frontier heuristic values are 8,4,6, greedy best-first selects the node with $h=4$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

11. A* Search

Definition: A* uses $f(n)=g(n)+h(n)$, combining actual cost so far with estimated remaining cost.

In simple words: $g(n)$ is path cost from start; $h(n)$ is estimated remaining cost. An admissible heuristic does not overestimate the true remaining cost. A consistent heuristic satisfies $h(n) \leq c(n, n') + h(n')$.

Detailed explanation: If $g(B)=5$ and $h(B)=3$ then $f(B)=8$. A* selects the lowest f frontier node under the algorithm's rules.

Example: Algorithm: initialize start; repeatedly select lowest f ; goal test; expand; update cheaper paths. A* can be optimal under the standard admissibility/consistency conditions, but may use substantial memory.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

12. Constraint Satisfaction Problem (CSP)

Definition: A CSP contains variables, domains and constraints.

In simple words: The task is to assign a value to every variable without violating constraints. Map coloring, Sudoku, N-Queens and scheduling are classic examples. Backtracking assigns values incrementally and returns when a partial assignment becomes invalid.

Detailed explanation: Map coloring: variables=regions, domain={red,green,blue}, constraint=adjacent regions must differ.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

13. Means-End Analysis

Definition: Means-end analysis compares the current state with a goal, identifies a difference and selects an operator that reduces the difference.

In simple words: Operators have preconditions and effects; unmet preconditions become subgoals.

Detailed explanation: To reach a destination, select an action that reduces distance; if a bridge is required, first create the subgoal of reaching it.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

14. Game Playing in AI

Definition: Game playing represents competitive decisions as a state space with alternating players.

In simple words: A game tree contains possible moves. Two-player zero-sum games are classic minimax settings. Terminal states receive utility values; large trees use depth limits and evaluation functions.

Detailed explanation: Chess: state=board plus player to move; action=legal move; utility=win/draw/loss.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

15. Minimax

Definition: Minimax is an adversarial decision rule for two-player zero-sum games: MAX maximizes and MIN minimizes.

In simple words: At MAX choose the largest child value; at MIN choose the smallest. The opponent is assumed to respond optimally.

Detailed explanation: If a MAX node has values 3,5,2 it returns 5; a MIN node with the same values returns 2.

Example: Typical time $O(b^m)$; depth-first space $O(bm)$, where b is branching factor and m is search depth.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

16. Alpha-Beta Pruning

Definition: Alpha-beta pruning removes branches that cannot change the minimax result.

In simple words: Alpha is MAX's best guaranteed value so far; beta is MIN's best guaranteed value so far. Prune when $\alpha \geq \beta$.

Detailed explanation: With good move ordering, many branches can be avoided while producing the same minimax decision.

Example: Worst case remains $O(b^m)$; ideal ordering is often described as $O(b^{(m/2)})$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

Unit 1 — Comparison Table

Method	Main idea	Key property
BFS	Shallowest first	Queue; equal-cost optimal
DFS	Deepest first	Stack/recursion; low memory
Greedy best-first	Lowest $h(n)$	Not generally optimal
A*	Lowest $g+h$	Optimal under standard conditions
Hill climbing	Local improvement	Can get stuck
Minimax	MAX vs MIN	Adversarial search
Alpha-beta	Prunes minimax	Same result, less work

UNIT 2 — KNOWLEDGE REPRESENTATION

This unit is expanded from every topic listed in the supplied syllabus. Each topic is presented with definition, simple explanation, working or key structure, example, exam point and relevant applications.

1. Knowledge Representation

Definition: Knowledge representation encodes facts, concepts, relationships and rules in a form an AI system can reason with.

In simple words: A knowledge base stores represented knowledge; an inference mechanism derives conclusions. Good representations balance expressiveness and efficient inference.

Detailed explanation: Fact: Student(Ali). Rule: Student(x)→Studies(x,AI).

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

2. Propositional Logic

Definition: Propositional logic represents statements that are true or false and combines them with logical connectives.

In simple words: Operators include NOT, AND, OR, implication and biconditional. Tautology is always true; contradiction always false; contingency depends on interpretation.

Detailed explanation: P =It rains, Q =Road is wet; $P \rightarrow Q$ means if it rains then the road is wet.

Example: Truth-table idea: $P \wedge Q$ is true only when both are true; $P \vee Q$ is false only when both are false; $P \rightarrow Q$ is false only when P is true and Q is false.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

3. First-Order Predicate Logic

Definition: FOL extends propositional logic with objects, predicates, variables, functions and quantifiers.

In simple words: Constants name objects; predicates represent properties/relations; $\forall$ means for all; $\exists$ means there exists.

Detailed explanation: $\forall x(\text{Student}(x) \rightarrow \text{Studies}(x, \text{AI}))$; $\exists x \text{ Student}(x)$.

Example: Compared with propositional logic, FOL is more expressive but inference is more complex.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

4. Resolution Principle

Definition: Resolution is an inference rule that combines clauses containing complementary literals.

In simple words: It is used in automated theorem proving, often with proof by contradiction: add the negated goal, convert to clauses, resolve until the empty clause or no new information.

Detailed explanation: $(P \vee Q)$ and $(\neg Q \vee R)$ resolve on Q to produce $(P \vee R)$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

5. Unification

Definition: Unification finds substitutions that make two expressions identical.

In simple words: It asks what variable values make two logical expressions match. An MGU is a most general unifier.

Detailed explanation: Unify Likes(x,Book) and Likes(Ali,y): {x/Ali,y/Book}.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

6. Semantic Networks

Definition: A semantic network is a graph where nodes represent concepts/objects and links represent relationships.

In simple words: Links can express IS-A, HAS-PART, OWNS and similar relations. Inheritance lets a specific concept receive properties from a general one.

Detailed explanation: Animal→IS-A→Bird→IS-A→Sparrow; Bird→HAS-PART→Wing.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

7. Conceptual Dependencies

Definition: Conceptual dependency represents underlying meaning using conceptual primitives and relations.

In simple words: It tries to capture what happened rather than merely preserve exact words, identifying participants and roles in an event.

Detailed explanation: A transfer-of-possession representation can contain actor, object and recipient even when wording changes.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

8. Frames

Definition: A frame is a slot-value structure representing an object, concept or situation.

In simple words: Slots can contain values, defaults and inherited information.

Detailed explanation: STUDENT frame: name, course, semester, university.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

9. Scripts

Definition: A script represents a stereotyped sequence of events.

In simple words: It is a story template for familiar situations. Components can include roles, props, scenes, entry conditions and results.

Detailed explanation: Restaurant: enter→seated→order→eat→bill→pay→leave.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

10. Production Rules

Definition: A production rule is an IF-THEN rule representing a condition and its consequence.

In simple words: Forward chaining starts from facts and derives consequences; backward chaining starts from a goal and searches for supporting rules.

Detailed explanation: IF temperature is high AND fever is present THEN consider a fever-related conclusion.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

Unit 2 — Comparison Table

Technique	Main strength
Propositional logic	Formal true/false reasoning
FOL	Objects, relations, quantifiers
Semantic network	Graph relationships
Frames	Slots, defaults, inheritance
Scripts	Stereotyped event sequences
Production rules	IF-THEN knowledge

UNIT 3 — UNDERSTANDING NATURAL LANGUAGES I

This unit is expanded from every topic listed in the supplied syllabus. Each topic is presented with definition, simple explanation, working or key structure, example, exam point and relevant applications.

1. Natural Language Processing

Definition: NLP is the AI field concerned with computational processing and understanding of human language.

In simple words: It can include lexical, syntactic, semantic and contextual processing.

Detailed explanation: Applications include translation, search, chatbots, information extraction, sentiment analysis and question answering.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

2. Components and Steps of Communication

Definition: Communication involves a sender, message, medium, receiver and interpretation.

In simple words: Language processing may involve words/lexicon, syntax, semantics and context.

Detailed explanation: Sender→message/language→medium→receiver→interpretation.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

3. Formal vs Natural Language

Definition: Formal language is explicitly defined by symbols and rules; natural language develops through human use and contains ambiguity and context dependence.

In simple words: Programming languages are formal; English and Hindi are natural.

Detailed explanation: Formal languages prioritize precision; natural languages prioritize flexible human communication.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

4. Grammar

Definition: Grammar is a set of rules describing how linguistic units combine into valid expressions.

In simple words: Grammar supports parsing and describes categories, order and recursive structures.

Detailed explanation: $S \rightarrow NP VP$; $NP \rightarrow Det N$; $VP \rightarrow V NP$ is a simplified phrase grammar.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

5. Chomsky Hierarchy

Definition: The hierarchy classifies grammars as Type 0 unrestricted, Type 1 context-sensitive, Type 2 context-free and Type 3 regular.

In simple words: It moves from more general to more restricted grammar classes. Associated machines are Turing machine, linear bounded automaton, pushdown automaton and finite automaton respectively.

Detailed explanation: Remember: Type 0 > Type 1 > Type 2 > Type 3 in generality.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

6. Parsing

Definition: Parsing determines syntactic structure from an input using grammar rules.

In simple words: A parser can produce a parse tree. Top-down begins with the start symbol; bottom-up begins with input symbols and builds upward.

Detailed explanation: Sentence $\rightarrow$ NP+VP is a simple parse-tree idea.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

7. Semantics

Definition: Semantics concerns the meaning of linguistic expressions.

In simple words: Syntax describes structure; semantics describes meaning. Context may affect interpretation.

Detailed explanation: “The student reads a book” has an agent-action-object interpretation.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

Chomsky Hierarchy — Memorize

Type	Grammar	Machine
0	Unrestricted	Turing machine
1	Context-sensitive	Linear bounded automaton
2	Context-free	Pushdown automaton
3	Regular	Finite automaton

UNIT 4 — UNDERSTANDING NATURAL LANGUAGES II

This unit is expanded from every topic listed in the supplied syllabus. Each topic is presented with definition, simple explanation, working or key structure, example, exam point and relevant applications.

1. Context-Free Grammar (CFG)

Definition: A CFG uses non-terminals, terminals, production rules and a start symbol; each production replaces one non-terminal by a sequence of symbols.

In simple words: A standard notation is $G=(V,\Sigma,P,S)$. Derivation repeatedly applies rules until terminals remain.

Detailed explanation: $S \rightarrow NP VP$; $NP \rightarrow Det N$; $VP \rightarrow V NP$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

2. Derivation and Parse Tree

Definition: A derivation is a sequence of production applications.

In simple words: Leftmost derivation expands the leftmost non-terminal first; rightmost expands the rightmost. A parse tree displays the hierarchy of the derivation.

Detailed explanation: Both derivation methods can generate the same sentence under a suitable grammar.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

3. CFG Applications in NLP

Definition: CFGs provide a foundation for phrase-structure syntax and parsing.

In simple words: They represent recursion and nested structures. Practical NLP often needs additional mechanisms for ambiguity and long-distance dependencies.

Detailed explanation: CFG can represent noun phrases, verb phrases and recursive clauses.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

4. Transformational Grammar

Definition: Transformational grammar relates underlying/deep structure and surface forms through transformations.

In simple words: It is used to describe relationships such as active/passive variations in traditional linguistic theories.

Detailed explanation: “The student solved the problem” and “The problem was solved by the student” have related event meanings but different surface structures.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

5. Recursive Transition Networks (RTN)

Definition: RTN represents grammar as states and transitions and can call sub-networks recursively.

In simple words: It behaves like a reusable grammar flowchart.

Detailed explanation: A sentence network can call NP and VP networks; recursion permits embedded structures.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

6. Augmented Transition Networks (ATN)

Definition: ATN extends RTN with registers, feature information, tests and actions.

In simple words: It can enforce constraints such as agreement and maintain information while parsing.

Detailed explanation: An ATN can store singular/plural number and test verb agreement.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

RTN vs ATN

Feature	RTN	ATN
Structure	Transition network with recursion	RTN plus augmentations
Memory	Basic	Registers/features
Conditions	Simple transitions	Tests/actions possible

UNIT 5 — AI IN DECISION MAKING

This unit is expanded from every topic listed in the supplied syllabus. Each topic is presented with definition, simple explanation, working or key structure, example, exam point and relevant applications.

1. Decision Making in AI

Definition: Decision making selects actions among alternatives using goals, preferences, evidence and uncertainty.

In simple words: Rules handle simple choices; utility and probability support uncertain decisions; sequential models consider future effects.

Detailed explanation: A route decision can trade time, cost and safety.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

2. Utility Theory

Definition: Utility theory represents preferences over outcomes using numerical utility.

In simple words: Higher utility means a more preferred outcome under the selected representation. Utility need not equal money; it can combine multiple objectives.

Detailed explanation: A safe but slower route may have higher utility than a risky faster route.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

3. Utility Functions

Definition: A utility function maps outcomes/states to utility values.

In simple words: Under uncertainty, expected utility weights utilities by probabilities.

Detailed explanation: $EU(A) = 0.8 \times 10 + 0.2 \times 0 = 8$.

Example: $EU(a) = \sum P(s|a)U(s)$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

4. Decision Networks

Definition: Decision networks combine chance nodes, decision nodes and utility nodes.

In simple words: Chance nodes represent uncertainty; decision nodes choices; utility nodes preferences.

Detailed explanation: Travel planning can include weather as a chance node, transport as a decision node and time/cost/comfort as utility.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

5. Sequential Decision Problems

Definition: A sequential decision problem contains multiple actions where current choices affect future states and rewards.

In simple words: The agent needs a policy rather than a single isolated choice.

Detailed explanation: Robot navigation is sequential because each movement changes the next state.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

6. Markov Decision Process (MDP)

Definition: An MDP models sequential decisions with states, actions, transition probabilities, rewards and a discount factor.

In simple words: The Markov property says the current state contains the information needed for predicting the future conditional on the current action.

Detailed explanation: Robot location=state; movement=action; transition=probability of next location; reward=delivery or penalty.

Example: $V\pi(s) = E\pi[\sum \gamma^t R_{t+1} | S_t = s]$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

7. POMDP

Definition: A POMDP extends an MDP when the true state is not directly observable.

In simple words: The agent receives observations and maintains a belief distribution over possible states.

Detailed explanation: A robot with noisy sensors may assign probabilities to several possible object locations.

Example: Core components: states, actions, transitions, observations, observation model, rewards and discount factor.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

8. Game Theory

Definition: Game theory studies strategic choices among multiple agents whose payoffs depend on everyone's actions.

In simple words: Players choose strategies while anticipating others. Concepts include payoffs, best responses, dominant strategies and equilibrium.

Detailed explanation: Two firms choosing prices can be represented by strategies and payoff values.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

9. Nash Equilibrium

Definition: A Nash equilibrium is a strategy profile in which no player can improve payoff by changing strategy alone while others stay fixed.

In simple words: It describes strategic stability, not necessarily the best collective outcome.

Detailed explanation: In a payoff matrix, look for a pair where each selected strategy is a best response to the other.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

10. Real-Time ML Decision Making

Definition: Machine learning can provide predictions, classifications and recommendations from incoming data to support rapid decisions.

In simple words: Typical pipeline: data validation→preprocessing→model inference→decision rule→action→monitoring.

Detailed explanation: Fraud detection can score a transaction immediately and trigger additional verification.

Example: Important concerns: latency, drift, data quality, safety, monitoring and fallback behavior.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

Decision Network Nodes

Node	Meaning
Chance	Uncertain event/variable
Decision	Choice made by agent
Utility	Preference/value

UNIT 6 — AI IN REINFORCEMENT LEARNING

This unit is expanded from every topic listed in the supplied syllabus. Each topic is presented with definition, simple explanation, working or key structure, example, exam point and relevant applications.

1. Reinforcement Learning

Definition: RL learns behavior by interaction: an agent observes a state, takes an action, receives reward and reaches a next state.

In simple words: Unlike supervised learning, the agent is not simply given a correct label for every situation; rewards can be delayed.

Detailed explanation: Game agent: state=board, action=move, reward=win/loss/score.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

2. Core RL Terms

Definition: State describes the situation; action is a choice; reward is feedback; policy selects actions; value estimates long-term return; Q-value estimates state-action return.

In simple words: The return is cumulative discounted reward.

Detailed explanation: $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$

Example: A policy can be deterministic or stochastic.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

3. Exploration vs Exploitation

Definition: Exploration tries uncertain actions to learn; exploitation uses actions currently believed to be best.

In simple words: The agent must balance discovery and use.

Detailed explanation: ϵ -greedy explores with probability ϵ and exploits with probability $1-\epsilon$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

4. Passive Reinforcement Learning

Definition: Passive RL evaluates a fixed policy rather than freely changing the policy.

In simple words: The agent follows the given policy and learns value estimates from experience.

Detailed explanation: A robot can evaluate a fixed navigation policy by recording rewards and returns.

Example: Process: follow policy → observe transitions/rewards → estimate values → repeat.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

5. Active Reinforcement Learning

Definition: Active RL lets the agent choose actions while learning what works.

In simple words: It must handle exploration/exploitation.

Detailed explanation: Q-learning is a classic model-free active method.

Example: $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

6. Generalization in Reinforcement Learning

Definition: Generalization means applying learned behavior to new but related states/situations rather than memorizing every state.

In simple words: Function approximation with features or neural networks is important in large or continuous spaces.

Detailed explanation: A robot trained on several layouts can transfer useful movement patterns to a new layout.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

7. Policy Search

Definition: Policy search directly optimizes a parameterized policy.

In simple words: A policy π_{θ} maps states/observations to actions; training adjusts θ to increase expected return.

Detailed explanation: A neural policy can map robot sensor values to control commands.

Example: $J(\theta) = E \pi_{\theta}[G]$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

8. Deep Reinforcement Learning

Definition: Deep RL uses neural networks to approximate policies, value functions or Q-values.

In simple words: DQN approximates Q-values; experience replay stores transitions; a target network stabilizes targets.

Detailed explanation: State/observation → network → Q-values → action; transitions are stored and sampled for training.

Example: DQN target for a non-terminal transition: $y = r + \gamma \max_{a'} Q_{\text{target}}(s', a')$.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

9. Real-Time RL Applications

Definition: RL can optimize repeated actions where feedback is available.

In simple words: Applications require safety constraints, reward design, simulation/testing, monitoring and latency control.

Detailed explanation: Traffic signal control can treat traffic conditions as states, signal phases as actions and reduced delay as reward.

Example: Examples: robotics, autonomous systems, games, recommendation, resource allocation, traffic control and industrial optimization.

Exam point: Start a long answer with the definition, then explain the structure/working, give an example, and finish with advantages/limitations and applications where applicable.

Common confusion: Do not treat a simplified classroom model as identical to every real-world implementation. State assumptions such as equal step cost, heuristic admissibility or full observability when relevant.

Quick self-test: Define the topic in 2 lines, explain it in 5 lines, give one example, and write two exam questions about it.

Passive vs Active RL

Feature	Passive RL	Active RL
Policy	Fixed	Agent learns/chooses
Main task	Policy evaluation	Action selection + learning
Exploration	Usually not central	Important

COMPLETE EXAM PREPARATION

The supplied request asks for unit-wise definitions, formulas, algorithms, diagrams, comparisons, 2-mark/5-mark/10-mark questions, numerical practice, viva questions, last-minute revision and a final syllabus checklist. ■filecite■turn0file0■L625-L678■

Important Formula Sheet

Topic	Formula / key relation
A*	$f(n)=g(n)+h(n)$
Expected utility	$EU(a)=\sum P(s a)U(s)$
Return	$G_t=R_{t+1}+\gamma R_{t+2}+\gamma^2 R_{t+3}+\dots$
Q-learning	$Q \leftarrow Q + \alpha [r + \gamma \max_{Q'} Q - Q]$
ϵ -greedy	Explore ϵ ; exploit $1-\epsilon$
Alpha-beta	Prune when $\alpha \geq \beta$

Algorithm Templates

BFS

- 1 FIFO queue
- 2 dequeue
- 3 goal test
- 4 expand/enqueue
- 5 repeat

DFS

- 1 Stack/recursion
- 2 pop
- 3 goal test
- 4 expand/push
- 5 repeat

Hill Climbing

- 1 Start
- 2 evaluate neighbors
- 3 move to improvement
- 4 stop when no acceptable improvement

Best-First

- 1 Priority queue
- 2 select best evaluation
- 3 goal test

4 expand

A*

- 1 $g(\text{start})=0$
- 2 select minimum $f=g+h$
- 3 goal test
- 4 relax/update successors

Minimax

- 1 Evaluate leaves
- 2 MAX takes maximum
- 3 MIN takes minimum
- 4 select root action

Alpha-Beta

- 1 $\alpha=-\infty, \beta=+\infty$
- 2 update α/β
- 3 prune when $\alpha \geq \beta$

Q-learning

- 1 observe state
- 2 choose action
- 3 observe reward/next state
- 4 apply Q update

Important Diagrams

Environment → Sensors → Agent → Actuators → Environment
A* frontier ranked by $f=g+h$
MAX → MIN → MAX → terminal utility
Percept → Knowledge Base + Inference → Action
Input → lexical/syntax → semantics → interpretation
Type 0 → Type 1 → Type 2 → Type 3
State → Action → Reward + Next State
Agent → Action → Environment → Reward/Next State → Agent
State → Neural Network → Q-values → Action

Important 2-Mark Questions

Unit 1

- Define AI.
- What is an intelligent agent?
- What is PEAS?
- Define heuristic.
- What is BFS?
- What is DFS?
- Define local maximum.
- What is a plateau?
- Define admissible heuristic.
- Write A* evaluation function.
- What is CSP?
- Define means-end analysis.
- What is a game tree?
- What is minimax?
- What is alpha-beta pruning?

Unit 2

- Define knowledge representation.
- What is a knowledge base?
- Define proposition.
- What is tautology?
- What is contradiction?
- Define FOL.
- What are quantifiers?
- What is resolution?
- Define unification.
- What is MGU?
- Define semantic network.
- What is a frame?
- What is a script?
- Define production rule.

Unit 3

- Define NLP.

- What is natural language?
- Define formal language.
- What is grammar?
- Define parsing.
- What is semantics?
- What is a parse tree?
- State the Chomsky hierarchy.
- What is CFG?

Unit 4

- Define CFG.
- What is a terminal?
- What is a non-terminal?
- Define derivation.
- What is leftmost derivation?
- What is rightmost derivation?
- Define transformational grammar.
- What is RTN?
- What is ATN?

Unit 5

- Define utility.
- What is a utility function?
- What is a decision network?
- What is an MDP?
- What is reward?
- What is a policy?
- What is partial observability?
- Define belief state.
- What is game theory?
- Define Nash equilibrium.

Unit 6

- Define reinforcement learning.
- What is a state?
- What is an action?
- What is reward?

- Define policy.
- What is a value function?
- What is Q-value?
- Define exploration.
- Define exploitation.
- What is passive RL?
- What is active RL?
- What is DQN?
- What is experience replay?
- What is a target network?

Important 5-Mark Questions

Unit 1

- Explain types of intelligent agents.
- Explain task-environment properties.
- Explain BFS with algorithm and complexity.
- Explain DFS with algorithm and complexity.
- Explain hill climbing and its problems.
- Explain heuristic and best-first search.
- Explain A* and heuristic conditions.
- Explain CSP with example.
- Explain means-end analysis.
- Explain minimax.
- Explain alpha-beta pruning.

Unit 2

- Explain propositional logic.
- Explain FOL and components.
- Explain resolution.
- Explain unification and MGU.
- Explain semantic networks.
- Explain frames.
- Explain scripts.
- Explain production rules and chaining.

Unit 3

- Explain NLP and applications.

- Compare formal and natural languages.
- Explain grammar.
- Explain Chomsky hierarchy.
- Explain parsing.
- Compare top-down and bottom-up parsing.
- Explain semantics.

Unit 4

- Explain CFG.
- Explain leftmost/rightmost derivation.
- Explain CFG applications in NLP.
- Explain transformational grammar.
- Explain RTN.
- Explain ATN.
- Compare RTN and ATN.

Unit 5

- Explain utility theory.
- Explain utility functions numerically.
- Explain decision networks.
- Explain sequential decision problems.
- Explain MDP.
- Explain POMDP and belief state.
- Explain game theory.
- Explain ML for real-time decisions.

Unit 6

- Explain RL terminology.
- Explain exploration vs exploitation.
- Explain passive RL.
- Explain active RL.
- Explain generalization.
- Explain policy search.
- Explain DQN.
- Explain experience replay and target network.
- Discuss real-time RL applications.

Important 10-Mark / Long-Answer Questions

Unit 1

- 1 Explain AI, intelligent agents, PEAS and task-environment properties.
- 2 Compare BFS, DFS, hill climbing, best-first and A*.
- 3 Explain A* with numerical example and heuristic conditions.
- 4 Explain CSP and backtracking using map coloring/N-Queens.
- 5 Explain minimax and alpha-beta pruning using a game tree.

Unit 2

- 1 Explain propositional and first-order predicate logic with comparison.
- 2 Explain resolution and unification step-by-step.
- 3 Discuss semantic networks, conceptual dependencies, frames, scripts and production rules.

Unit 3

- 1 Explain NLP, communication, formal vs natural language, grammar, parsing and semantics.
- 2 Explain Chomsky hierarchy with grammar types and associated automata.
- 3 Explain top-down and bottom-up parsing.

Unit 4

- 1 Explain CFG, components, derivation and parse tree.
- 2 Explain transformational grammar with deep/surface structure.
- 3 Explain RTN and ATN and compare them.

Unit 5

- 1 Explain utility theory, utility functions and decision networks.
- 2 Explain MDP and POMDP with components and belief states.
- 3 Explain game theory and Nash equilibrium in AI.
- 4 Discuss ML for real-time decision making.

Unit 6

- 1 Explain RL and all core terms.
- 2 Compare passive and active RL.
- 3 Explain exploration/exploitation and Q-learning.
- 4 Explain deep RL and DQN including replay and target network.
- 5 Discuss real-time RL applications.

Numerical / Problem-Solving Practice**A***

Edges $S-A=2$, $S-B=5$, $A-C=2$, $B-C=1$, $C-G=3$. Heuristics $h(S)=6, h(A)=5, h(B)=3, h(C)=3, h(G)=0$. Compute f values and trace A*.

BFS/DFS

Tree $A \rightarrow B, C$; $B \rightarrow D, E$; $C \rightarrow F, G$; $E \rightarrow H$. Write BFS and a left-to-right DFS order.

Minimax

Root MAX has two MIN children: left leaves 3,5; right leaves 2,9. Compute the minimax value and chosen branch.

Expected utility

A gives utility 10 with probability .6 and utility 2 with .4; B gives utility 7 with probability 1. Compute both expected utilities.

Unification

Unify $\text{Likes}(x, \text{Book})$ with $\text{Likes}(\text{Ali}, y)$.

Q-learning

$Q=4$, $\alpha=.5$, $r=3$, $\gamma=.9$, $\max Q'=6$. Compute the updated Q .

Selected answers

- BFS: A,B,C,D,E,F,G,H. DFS depends on child order; left-to-right gives A,B,D,E,H,C,F,G.
- Minimax: left MIN=3, right MIN=2, root MAX=3.
- $EU(A) = .6 \times 10 + .4 \times 2 = 6.8$; $EU(B) = 7$.
- Unification: $\{x/\text{Ali}, y/\text{Book}\}$.
- $Q \text{ update} = 4 + .5[3 + .9(6) - 4] = 6.2$.

Viva Questions with Short Answers

Q. What is AI?

A. A field concerned with intelligent perception, reasoning, learning and action.

Q. What is an agent?

A. A system that perceives an environment and acts on it.

Q. What does PEAS mean?

A. Performance measure, Environment, Actuators, Sensors.

Q. Why is A* informed?

A. It uses heuristic information $h(n)$.

Q. What is $g(n)$?

A. Actual path cost from start to n .

Q. What is $h(n)$?

A. Estimated remaining cost.

Q. What is an admissible heuristic?

A. A heuristic that does not overestimate true remaining cost.

Q. What is a CSP?

A. Variables, domains and constraints.

Q. What is unification?

A. Finding substitutions that make expressions match.

Q. What is parsing?

A. Constructing syntactic structure from input.

Q. What is semantics?

A. Meaning of linguistic expressions.

Q. What is an MDP?

A. Sequential decision model with states, actions, transitions and rewards.

Q. What is a POMDP?

A. An MDP with partial observability and observations/beliefs.

Q. What is utility?

A. A numerical representation of preference.

Q. What is RL?

A. Learning through interaction and reward.

Q. What is a policy?

A. A mapping from states/observations to actions.

Q. What is Q-learning?

A. A model-free RL method that updates action values from reward and estimated future value.

Q. What is DQN?

A. A deep neural-network method for approximating Q-values.

Last-Minute Revision Sheets

Unit 1

- AI includes search, reasoning, learning and action.
- Agent=sensors+decision+actuators.
- PEAS=Performance, Environment, Actuators, Sensors.
- BFS=queue/shallowest; DFS=stack/deepest.
- Hill climbing can face local maxima, plateaus and ridges.
- Best-first often uses h ; A* uses $g+h$.
- CSP=variables+domains+constraints.
- Minimax=MAX/minimize opponent; alpha-beta prunes when $\alpha \geq \beta$.

Unit 2

- Propositional logic uses truth-valued propositions.
- FOL adds objects, relations and quantifiers.
- Resolution combines clauses.
- Unification finds substitutions.
- Semantic network=nodes+links.
- Frame=slots+values.
- Script=event sequence.
- Production rule=IF-THEN.

Unit 3

- NLP processes human language.
- Formal language is rule-controlled; natural language is flexible and ambiguous.
- Grammar defines structures.
- Chomsky: 0 unrestricted, 1 context-sensitive, 2 context-free, 3 regular.
- Parsing builds syntax; semantics concerns meaning.

Unit 4

- CFG uses terminals, non-terminals, productions and start symbol.
- Leftmost/rightmost derivations differ by expansion order.
- RTN uses recursive transition networks.
- ATN adds memory/features/tests/actions.

Unit 5

- Utility represents preference.
- Expected utility weights outcomes by probability.
- Decision network=chance+decision+utility.

- MDP=state+action+transition+reward+discount.
- POMDP adds observations and belief states.
- Nash equilibrium=no profitable unilateral deviation.

Unit 6

- RL=state→action→reward/next state.
- Policy selects actions; value estimates return; Q estimates state-action return.
- Explore=learn; exploit=use known best.
- Passive evaluates fixed policy; active learns while acting.
- Q-learning update uses α , γ and max future Q.
- DQN uses neural network, replay and target network.

Final Syllabus Checklist

- Unit 1 complete: AI, types, agents, BFS, DFS, hill climbing, heuristic search, best-first, A*, CSP, means-end, games, minimax, alpha-beta.
- Unit 2 complete: knowledge representation, propositional logic, FOL, resolution, unification, semantic networks, conceptual dependencies, frames, scripts, production rules.
- Unit 3 complete: communication, formal/natural language, grammar, Chomsky hierarchy, parsing, semantics.
- Unit 4 complete: CFG, derivation, parse tree, transformational grammar, RTN, ATN.
- Unit 5 complete: utility, utility functions, decision networks, sequential decisions, MDP/POMDP, game theory, real-time ML.
- Unit 6 complete: RL, passive/active RL, generalization, policy search, deep RL/DQN, real-time RL applications.
- Exam section complete: definitions, formulas, algorithms, diagrams, comparisons, 2/5/10-mark questions, numericals, viva and revision.

Final Study Strategy

- 1 First pass: learn every definition and simple explanation.
- 2 Second pass: reproduce algorithms, diagrams and comparison tables.
- 3 Third pass: solve numerical and logic/search problems.
- 4 Fourth pass: write timed 2-mark, 5-mark and 10-mark answers.
- 5 Final revision: use the last-minute sheets and checklist.